

REQUIREMENT SPECIFICATION DOCUMENT

COLLEGE STUDENT MANAGEMENT DATABASE SYSTEM

Week 1 – Requirement Analysis and Student Database Module

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1. INTRODUCTION

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the College Student Management Database System. This system is designed to manage and organize important college information such as student details, department details, course details, faculty information, attendance, marks, and academic records.

In many colleges, student information is maintained using paper records, registers, and separate files. This method may take more time and can lead to problems such as missing records, duplicate entries, incorrect information, and difficulty in finding student details.

The College Student Management Database System helps to reduce these problems by storing all student-related information in a centralized and organized database. The system will help college staff maintain accurate records and access student information quickly.

1.2 Project Scope

The College Student Management Database System will allow the college to store and manage information about students, departments, courses, subjects, faculty members, attendance, marks, and academic performance.

The system will support important college activities such as student registration, department allocation, course management, attendance recording, marks entry, and result generation.

The system will be developed using a relational database with properly defined tables and relationships. The database will help maintain accurate, secure, and consistent student information.

The system will also support the generation of reports such as student lists, department-wise student details, attendance reports, marks reports, and academic performance reports.

1.3 Intended Users

The following users will use the College Student Management Database System:

Students:

Students can view their personal details, attendance, marks, results, and academic information.

Faculty Members:

Faculty members can view student details, record attendance, enter marks, and manage subject-related information.

College Administrators:

Administrators can manage student records, departments, courses, faculty information, and generate reports.

Office Staff:

Office staff can register new students, update student details, and maintain academic records.

2. BUSINESS REQUIREMENTS

- ❖ The College Student Management Database System must satisfy the following business requirements:
- ❖ Maintain accurate and up-to-date records of all students.
- ❖ Store complete information about college departments and courses.
- ❖ Allow new students to register with their personal and academic details.
- ❖ Maintain student information such as name, register number, department, course, contact number, and email address.
- ❖ Allow administrators to add, update, and remove student records.
- ❖ Maintain information about faculty members and subjects.
- ❖ Record and manage student attendance.
- ❖ Store student marks and examination results.
- ❖ Calculate student attendance percentage.
- ❖ Generate student academic reports.
- ❖ Maintain department-wise and course-wise student records.

- ❖ Allow students to view their attendance and marks.
- ❖ Generate reports about students, attendance, marks, and academic performance.
- ❖ Reduce manual work and improve the efficiency of college administration.

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe the activities and operations that the system must perform.

ID	Requirement Description
FR-1	The system shall allow new students to register with their personal and academic details.
FR-2	The system shall generate a unique student identification number for every student.
FR-3	The system shall store student details such as name, register number, department, course, phone number, and email address.
FR-4	The system shall allow administrators to view and update student information.
FR-5	The system shall allow administrators to add, update, and remove department details.
FR-6	The system shall allow administrators to manage course and subject information.
FR-7	The system shall store faculty details such as faculty name, department, and subject.
FR-8	The system shall allow faculty members to record daily student attendance.
FR-9	The system shall calculate the attendance percentage of each student.
FR-10	The system shall allow faculty members to enter student examination marks.
FR-11	The system shall calculate student results based on marks.
FR-12	The system shall allow students to view their attendance and marks.
FR-13	The system shall maintain the academic records of every student.
FR-14	The system shall generate department-wise student reports.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality and performance standards of the system.

4.1 Performance

The system should retrieve student information quickly. Student records, attendance details, and marks should be displayed within a few seconds. The system should process student registration and record updates without unnecessary delay.

4.2 Security

The system must protect student information. Only authorized college administrators, faculty members, and office staff should be allowed to access or modify student records. Student login information should be stored securely.

4.3 Reliability

The system should maintain accurate and consistent student records. Attendance, marks, and academic information should be stored correctly. The system should prevent duplicate and incorrect student records.

4.4 Usability

The system should have a simple and user-friendly interface. Students, faculty members, and college staff should be able to use the system easily. The forms and options should be clear and easy to understand.

4.5 Scalability

The database should support an increasing number of students, departments, courses, subjects, and academic records. The system should continue to work efficiently when more information is added in the future.

4.6 Maintainability

The system should be easy to maintain and update. New departments, courses, subjects, and features should be added without affecting the existing database records.

4.7 Data Integrity

The system must maintain correct relationships between students, departments, courses, subjects, attendance, and marks. Each student record should be connected to a valid department and course.

5. STUDENT DATABASE MODULE – DATA REQUIREMENTS

The Student Database Module is an important part of the College Student Management Database System. It stores complete information about all students studying in the college.

The following attributes are identified for the Student Database Module:

Attribute	Data Type	Description
student_id	INT (Primary Key)	Unique identification number for each student
full_name	VARCHAR(100)	Full name of the student
register_number	VARCHAR(30)	Unique register number of the student
date_of_birth	DATE	Date of birth of the student
gender	VARCHAR(20)	Gender of the student
department	VARCHAR(100)	Department in which the student studies
course	VARCHAR(100)	Course selected by the student
year_of_study	INT	Current year of study
phone_number	VARCHAR(15)	Contact number of the student
email	VARCHAR(100)	Email address of the student
address	VARCHAR(255)	Residential address of the student
admission_date	DATE	Date on which the student joined the college

The student_id will be used as the primary key because it uniquely identifies every student record. The register number will also be used to identify students in the college.

6. DEPARTMENT AND COURSE DATABASE MODULE

The Department and Course Database Module stores information about college departments and courses.

The following attributes are identified:

7. ATTENDANCE AND MARK DATABASE MODULE

The Attendance Module records the daily attendance of students.

The following attributes are identified:

Attribute	Data Type	Description
attendance_id	INT (Primary Key)	Unique identification number for each attendance record
student_id	INT (Foreign Key)	Identifies the student
subject_name	VARCHAR(100)	Name of the subject
attendance_date	DATE	Date on which attendance was recorded
attendance_status	VARCHAR(20)	Status such as Present or Absent

The student_id is a foreign key because it connects the attendance record with the Student Database Module.

The Marks Module stores student examination marks and academic results.

Attribute	Data Type	Description
mark_id	INT (Primary Key)	Unique identification number for each marks record
student_id	INT (Foreign Key)	Identifies the student
subject_name	VARCHAR(100)	Name of the subject
internal_mark	INT	Internal examination mark
external_mark	INT	External examination mark
total_mark	INT	Total mark obtained by the student
result	VARCHAR(20)	Result such as Pass or Fail

8. ASSUMPTIONS AND CONSTRAINTS

The following assumptions and constraints are considered for the College Student Management Database System:

- 1) The system will be developed using a relational database such as MySQL, Oracle, or PostgreSQL.
- 2) Every student will have a unique student identification number.
- 3) Every student will have a unique register number.
- 4) Only authorized college staff will be allowed to add, update, or delete student records.
- 5) Each student must be connected to a valid department and course.
- 6) Faculty members will be responsible for recording student attendance and marks.
- 7) Student information must be stored securely.
- 8) The system should prevent duplicate student records.
- 9) Attendance percentage should be calculated correctly.
- 10) Student marks and results should be maintained accurately.
- 11) The system should maintain correct relationships between students, departments, courses, attendance, and marks.
- 12) This document mainly focuses on the initial requirements and database modules of the College Student Management Database System.

9. CONCLUSION

The College Student Management Database System will help improve the efficiency and accuracy of college administration. The system will provide a centralized database for storing information about students, departments, courses, faculty members, attendance, marks, and academic records.

The proposed system will reduce manual work and make it easier for college staff to manage a large number of student records. It will also help faculty members maintain attendance and marks efficiently.

The requirement specification document identifies the important business requirements, functional requirements, non-functional requirements, and data requirements of the system. The Student Database Module, Department and Course Module, Attendance Module, and Marks Module provide the foundation for designing the database.

This document will be useful for creating the Entity Relationship Diagram (ER Diagram), designing database tables, defining primary and foreign keys, and implementing the College Student Management Database System in future development stages.